

CS402/MA492 Revision & Exam Preparation

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Generalities

- Your exam will be a timed **open book exam** on Blackboard.
- You will have **two hours** to work on the exam and an additional 30 minutes to upload your solutions.
- You should submit your solutions as a **single PDF file**.
NUI Galway recommends that you use Microsoft **Office Lens** to produce a legible (!) scan.
- You may be interviewed if concerns in relation to the integrity of your exam should arise.

Content

- The exam paper will be similar in content, form, and length to previous years, in particular to last year's paper.
- There will be both **theoretical** and **practical** questions.
- Have a look at past papers for sample questions.
You can find links on the course web page.
- There will be no questions that ask you to read or write Python code. You will not be expected to use Python to answer any of the practical questions. A pocket calculator will suffice.

Next

We'll now go through some of the practical questions from the 2018/2019 Cryptography exam paper.

Q1(a)

Problem

The ciphertext string

ASISENGEHFGNONFVWTGNQWCVSE

was produced using a shift cipher over the 27-letter alphabet

$$A = 0, B = 1, \dots, Z = 25, _ = 26,$$

where the final character $_$ indicates a space.

By conducting a frequency analysis of a larger sample of ciphertexts, it has been found that the plaintext symbol E corresponds to the ciphertext symbol S.

Determine the first word of the plaintext string corresponding to the ciphertext string given above.

Q1(a) (contd)

Solution:

A	B	C	D	E	F	G	H	I	J	K	L	M	N
0	1	2	3	4	5	6	7	8	9	10	11	12	13
O	P	Q	R	S	T	U	V	W	X	Y	Z	␣	
14	15	16	17	18	19	20	21	22	23	24	25	26	

- Encryption: $x \mapsto (x + k) \bmod 27$ for an unknown key $k \in \mathbb{Z}_{27}$.
- The plaintext E ($\leftrightarrow 4$) corresponds to the ciphertext S ($\leftrightarrow 18$).
- Hence, $k = 14$ and decryption is

$$y \mapsto (y - 14) \bmod 27 = (y + 13) \bmod 27.$$

- Ciphertext string: ASISENGEHFGNONFVWTGNQWCVSE
- Decryption:

ciphertext	y	$(x + 13) \bmod 27$	plaintext
A	0	13	N
S	18	4	E
I	8	21	V
E	4	17	R
N	13	26	␣

Q1(a) (contd)

Hence, the first word of the plaintext string corresponding to the ciphertext string that we were given is

NEVER

Q1(b)

Problem

Identify $\mathbb{Z}_{27} = \{0, \dots, 25, 26\}$ with $\{A, \dots, Z, \sqcup\}$ as in part (a). An affine cipher over $\mathbb{Z}_{27}$ with unknown key $(a, b) \in \mathbb{Z}_{27}^\times \times \mathbb{Z}_{27}$ and encryption function

$$\mathbb{Z}_{27} \rightarrow \mathbb{Z}_{27}, \quad x \mapsto ax + b$$

encrypts E as T and S as V. Determine (a, b) .

Solution:

A	B	C	D	E	F	G	H	I	J	K	L	M	N
0	1	2	3	4	5	6	7	8	9	10	11	12	13
O	P	Q	R	S	T	U	V	W	X	Y	Z	$\sqcup$	
14	15	16	17	18	19	20	21	22	23	24	25	26	

- E $\rightsquigarrow$ T gives $4a + b \equiv 19 \pmod{27}$.
- S $\rightsquigarrow$ V gives $18a + b \equiv 21 \pmod{27}$.
- Hence, $14a \equiv 2 \pmod{27}$. 14 is invertible modulo 27 (because $14 = 2 \times 7$ and $27 = 3^3$) with inverse 2. Hence, $a \equiv 4 \pmod{27}$.
- $4^2 + b \equiv 19 \pmod{27}$ gives $b \equiv 3 \pmod{27}$. $\rightsquigarrow (a, b) = (4, 3)$.

Q2(d)

Problem

A linear feedback shift register of length 3 with connection polynomial

$$C(X) = 1 + X + X^3$$

over $\mathbb{Z}_2$ is used to produce a pseudo-random sequence of binary digits with initial values $s_0 = 0$, $s_1 = 0$, and $s_2 = 1$. Determine the least period of the resulting infinite sequence $s_0, s_1, \dots$.

Recall ($\rightarrow$ Lecture 11)

An LFSR of length L with connection polynomial

$$C(X) = 1 + c_1X + \dots + c_LX^L$$

has the feedback function

$$f(s_{j-1}, \dots, s_{j-L}) = c_1s_{j-1} + \dots + c_Ls_{j-L}.$$

Q2(d) (contd)

Solution:

- Connection polynomial: $C(X) = 1 + X + X^3$
- Initial state: $s_0 = 0, s_1 = 0, s_2 = 1$
- Hence,
 - ▶ $s_3 = f(s_2, s_1, s_0) = s_2 + s_0,$
 - ▶ $s_4 = f(s_3, s_2, s_1) = s_3 + s_1,$
 - ▶ ...

t	s_{t+2}	s_{t+1}	s_t	
0	1	0	0	← initial state
1	1	1	0	
2	1	1	1	
3	0	1	1	
4	1	0	1	
5	0	1	0	
6	0	0	1	
7	1	0	0	← initial state!

We conclude that the least period is 7.

Q3(c)

Problem

Alice and Bob use the cyclic group $\mathbb{Z}_{25}^\times$ and its generator $g = 2$ to perform a *Diffie-Hellman key exchange*. Alice secretly chooses $a = 6$ and Bob secretly chooses $b = 3$. Determine their shared key (as an element of $\mathbb{Z}_{25}$).

Solution:

$$g^{ab} = (g^a)^b = (2^6)^3 = 64^3 = 14^3 = \boxed{19} \text{ in } \mathbb{Z}_{25}^\times.$$

Note:

$$14^2 = 196 = -4 = 21 \text{ and } 14^3 = 14 \times 21 = 294 = -6 = 19, \text{ all in } \mathbb{Z}_{25}^\times.$$

Q3(d)

Problem

Alice encrypts the message $m = 23$ using the RSA cryptosystem and Bob's public key $(n, e) = (77, 17)$.

Find the corresponding ciphertext (as an element of $\mathbb{Z}_{77}$).

Reminder: RSA encryption and decryption

- To encrypt a message $m \in \mathbb{Z}_n$ using Alice's public key (n, e) , Bob computes

$$c = m^e \bmod n.$$

He then sends the ciphertext c to Alice.

- To decrypt c , Alice uses her private key (n, d) and computes

$$m = c^d \bmod n.$$

Q3(d) (contd)

Solution:

$$c = m^e = 23^{17} \bmod 77.$$

There are various ways of computing this number without a computer.

Let's use the method of **fast exponentiation** ($\rightarrow$ Lecture 18).

- Write 17 as a sum of powers of 2:

$$17 = 1 + 16 = 2^0 + 2^4.$$

- By repeatedly squaring, compute $23^{2^i} \bmod 77$ for $i \leq 4$:
 - ▶ $23^2 \bmod 77 = 529 \bmod 77 = 67,$
 - ▶ $23^4 \bmod 77 = 67^2 \bmod 77 = 23,$
 - ▶ $23^8 \bmod 77 = 23^2 \bmod 77 = 67,$
 - ▶ $23^{16} \bmod 77 = 67^2 \bmod 77 = 23.$
- $23^{17} \equiv 23 \times 23^{16} \equiv 23 \times 23 \equiv 67 \pmod{77}$ so that $c = 67$.